Building Your OE Library Files — Step by Step

You have a master list of metabolites for the QE1 instrument (`QE1_Lib_POS.csv` and its negative-mode twin). You ran a subset of those same standards on a *different* instrument (OE). You need to build a **new, smaller file** that has just the standards you actually ran, with **two retention time (rt) columns** — one from the old QE1 run, one from your new OE run — so people can compare them.

You'll end up with 4 files total: POS and NEG versions, for QE1-based and OE-based libraries.

Step 1 — Get the starting file

1. Go to: https://github.com/erflam/Lab/tree/main/Library%20Creation
2. Click on `QE1_Lib_POS.csv`
3. Click the **Download raw file** button (usually looks like a down-arrow icon) to save it to your computer — don't just copy/paste the on-screen text, download the actual file.
4. Put it in a folder you'll remember, e.g. `Desktop/Library Creation/`

Step 2 — Make a new empty file for your results

1. In that same folder, make a copy of `QE1_Lib_POS.csv`
2. Rename the copy to `OE_Lib_POS.csv` *(You'll delete out the rows you don't need in the next step — easier than typing a file from scratch.)*

Step 3 — Keep only the standards you actually ran

Open `OE_Lib_POS.csv` in Excel (or Google Sheets).

- Go through the `name` column.
- **Delete every row** for a metabolite you did *not* run as a standard.
- Keep every row for metabolites you *did* run — including all their variants (e.g. `ADENINE`, `ADENINE-NH3`, `ADENINE+K`, `ADENINE+Na` all stay if you ran adenine).

> 💡 Tip: If you have a lot of rows, it's faster to filter/sort by `name`, then delete whole blocks at once rather than hunting row by row.

Step 4 — Rename the existing rt column

Find the column currently called `rt` (this came from the QE1 file).

- Rename its header to **`rt QE`**
- This just means "this retention time number is from the QE1 run" — you're not changing any of the numbers, just relabeling the column so it's clear where it came from.

Step 5 — Add your new column

- Add a brand new column to the right of everything else, with the header **`rt OE`**
- For each row, type in the retention time **you personally observed** when you ran that standard on the OE instrument.
- This is the only column where you're typing in brand-new numbers — everything else already exists in the file.

Step 6 — Check it looks right

Your finished `OE_Lib_POS.csv` should look like this (example):

adduct	mz	rt QE	name	Formula	rt OE
M+H	195.07699	4.74	4-Aminohippuric Acid	C9H10N2O3	4.25
M+H	136.0618	1.48	ADENINE	C5H5N5	1.53
M+H-NH3	119.0352	1.48	ADENINE-NH3	C5H5N5	1.53

Make sure:

- ✅ Every standard you ran is represented (including all its adduct variants)
- ✅ `rt QE` has the *old* numbers, `rt OE` has your *new* numbers
- ✅ No extra/unrelated metabolites are left in the file
- ✅ Save it as a `.csv` file (not `.xlsx`)

Step 7 — Do the same thing for negative mode

Repeat Steps 1–6, but starting from the **negative mode** file instead (something like `QE1_Lib_NEG.csv` in the same GitHub folder), and name your new file `OE_Lib_NEG.csv`.

Step 8 — Confirm you have 4 files

By this point you should have:

1. `QE1_Lib_POS.csv` (original, downloaded)
2. `QE1_Lib_NEG.csv` (original, downloaded)
3. `OE_Lib_POS.csv` (your new file, built from the POS one)
4. `OE_Lib_NEG.csv` (your new file, built from the NEG one)

Step 9 — Download the processing code

Go back to the GitHub folder and download whatever script/code file is there (likely a `.py` file). Save it into the **same folder** as your 4 CSVs — that makes step 10 easier.

Step 10 — Point the code at your files

Open the code file in a text editor (or in Python's IDE / VS Code).

- Near the top, there will be lines that reference file names or file paths (something like `"QE1_Lib_POS.csv"` or a full path like `"C:/Users/you/Desktop/QE1_Lib_POS.csv"`).
- Replace those with the **actual location/name** of your files from Step 8. Easiest way: if the code and your CSVs are in the same folder, you can usually just use the plain filename (e.g. `"OE_Lib_POS.csv"`) instead of a long path.

Step 11 — Run it

- If it's a Python script: open a terminal/command prompt, navigate to that folder, and run `Library Adjustment.py` (using the code's actual filename).
- Check that it runs without errors and produces the expected output.

Quick glossary

- **rt** = retention time (how long the compound took to come off the instrument)
- **mz** = mass-to-charge ratio (identifies the compound's mass)
- **adduct** = the form the ion took (e.g. gained a proton = M+H, gained sodium = M+Na)
- **QE1 / OE** = the two different instruments being compared